DL FOR MUSIC GENERATION AND ANALYSIS HW1

2024 / 09 / 30

R12942148

電信丙 林峻佑

Source Code and Model Link

OUTLINE

1. **Methodology**
2. **Approach Details**
3. **Result Highlights**
4. **Result Analysis**
5. **Main Takeaway**
6. **Appendix**

METHODOLOGY

- Without fine-tuned (Vanilla)
  - Training only the classifier
- Fine-tuned model
  - Training both the feature extractor and the classifier
- Weighted loss
  - Since I have observed that the distribution of the labels are quite imbalanced, especially the percussion and piano, I also attempt to apply weighted loss to deal with it.
  - Weight score

    $weight_j = \frac{\sum_{i=1}^{N_j} \mathbb{I}[Y_{i,j} = 0]}{\sum_{i=1}^{N_j} \mathbb{I}[Y_{i,j} = 1]}$, where $i \in \{1, \dots, N_j\}$, $j \in \{0, \dots, 8\}$, N_j is the number of data of class j
- Fine-tuned & Weighted loss

APPROACH DETAILS – MERT

- Main Concept
  - Music shares lots of similarity with human language, making Pre-trained Language Model (PLM) could be adopted to deal with music sequence.
  - Multi-task training for learning the distinctiveness of the representations between acoustic and music.
- Pre-training
  - Similar to other language model (e.g. BERT), they use the concept of masked language modelling to perform self-supervise learning (reconstruct the waveform of the music sequence)
  - Using the MFCC features as the model input, but MFCC is only good at representing single-pitch signal, but not acoustic and timbre information
    → try to use other methods as the teacher to learn the acoustic information.

Source: MERT: ACOUSTIC MUSIC UNDERSTANDING MODEL WITH LARGE-SCALE SELF-SUPERVISED TRAINING

APPROACH DETAILS – MERT

- Acoustic learning
  - Teacher model: EnCodec
  - converts input waveforms to 8 different embeddings at 75Hz with a 320-fold reduction
  - Task: reconstruct the waveforms from the $L \times 8$ embeddings
- Music learning
  - Teacher: CQT
  - Task: reconstruct the CQT spectrum of the masked input (with MSE as the loss function)

$$\mathcal{L}_{CQT}(f_{cqt}; x, M, \mathbf{z}_{cqt}) = \sum_{t \in M} \|z_{cqt,t} - f_{cqt}(x')_t\|_2$$

RESULT HIGHLIGHTS

- In most metrics, fine-tuned without weighted loss has the best performances.
- If we combine weighted loss, the performance of macro average recall increases about 10%, while the weighted and micro average have been tied with the fine-tuned one.
- Even with only the weighted loss term, the recall value still have about 2% ~ 23% increasement in most cases.

	Vanilla			Weighted Loss			Fine-tuned			Fine-tuned + Weighted Loss		
	Precision	Recall	f1-Score	Precision	Recall	f1-Score	Precision	Recall	f1-Score	Precision	Recall	f1-Score
micro avg	0.87	0.71	0.78	0.65	0.73	0.69	**0.88**	**0.83**	**0.85**	0.83	**0.83**	0.83
macro avg	0.71	0.45	0.48	0.60	0.68	0.59	**0.84**	0.67	0.72	0.76	**0.77**	**0.76**
weighted avg	0.82	0.71	0.71	0.76	0.73	0.72	**0.88**	**0.83**	**0.84**	0.84	**0.83**	0.83
samples avg	0.86	0.72	0.77	0.62	0.70	0.64	**0.88**	**0.83**	**0.84**	0.81	0.80	0.79

RESULT ANALYSIS

- Classification report
  - For those minor classes, such as Percussion and Organ, the weighted loss do a great improvement on the f1-score, even beat down the fine-tuned model.
  - The Vanilla model even mis-predicts all the test samples of the Percussion class
  - However, the fine-tuned model still outperform in most metrics except the recall values.

	Vanilla			Weighted Loss			Fine-tuned			Fine-tuned + Weighted Loss			
Class	Precision	Recall	f1-Score	Precision	Recall	f1-Score	Precision	Recall	f1-Score	Precision	Recall	f1-Score	Supports
Piano	0.86	0.99	0.92	0.87	0.66	0.75	0.91	0.97	0.94	0.89	0.85	0.87	1889
Percussion	0	0	0	0.2	0.46	0.27	0.81	0.16	0.26	0.45	0.64	0.53	243
Organ	0.6	0.07	0.12	0.24	0.93	0.38	0.72	0.45	0.55	0.64	0.7	0.67	461
Guitar	0.88	0.99	0.93	0.89	0.91	0.9	0.91	0.97	0.94	0.92	0.87	0.89	1943
Bass	0.94	0.99	0.96	0.95	0.88	0.91	0.98	0.98	0.98	0.98	0.93	0.95	2076
Strings	0.85	0.63	0.72	0.85	0.57	0.68	0.95	0.69	0.8	0.85	0.8	0.83	1235
Voice	0.82	0.07	0.13	0.4	0.68	0.5	0.9	0.51	0.65	0.77	0.72	0.74	485
Wind Instruments	0.66	0.32	0.43	0.45	0.68	0.54	0.66	0.83	0.73	0.64	0.83	0.72	889
Synths	0.73	0.03	0.06	0.53	0.3	0.38	0.71	0.52	0.6	0.66	0.58	0.62	647

RESULT ANALYSIS

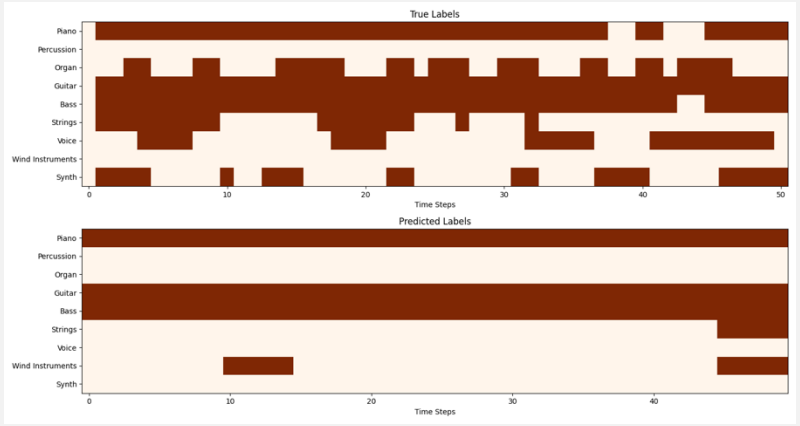
- Piano Roll Prediction
  - I choose the Track01876 and Track02100 for the following analysis, since the Percussion and Organ have appeared in these two tracks. (the figure can be found in the appendix)
- Track01876
  - Only the fine-tuned + weighted loss has correctly identified the Synth label.
  - For the Organ label, only the two weighted loss models have identified in the correct period.
  - However, the two without fine-tuned have higher probability to mis-recognized the sounds, such as the wind instruments, which has never appeared in the whole sound track.
- Track 02100
  - Only the two models with weighted loss have identified the Percussion sound, though the model without fine-tuned just simply predict the Percussion has been appearing most of the time.
  - The fine-tuned model without weighted loss correctly recognized sounds near the whole track duration.

MAIN TAKEAWAY

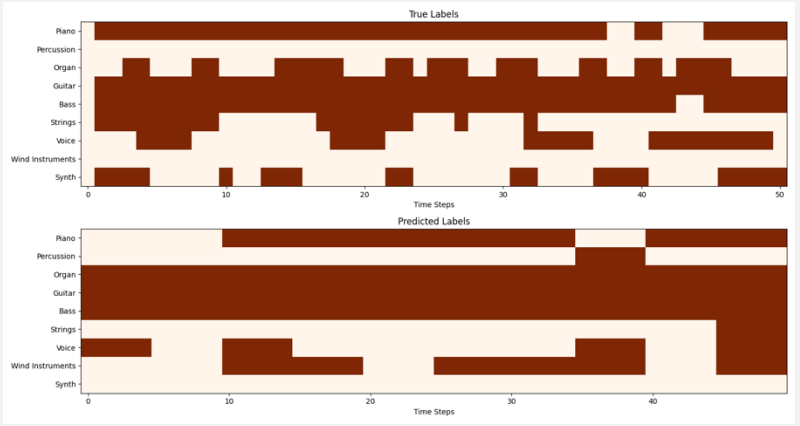
- The fine-tuned model has better performance to the one without fine-tuned.
- Although weighted loss do a great influence to make the model put more emphasis on the minor classes, it will also hurt the accuracy on the majors.
- The f1-score of fine-tuned model without weighted loss can achieve to about 84%.

APPENDIX

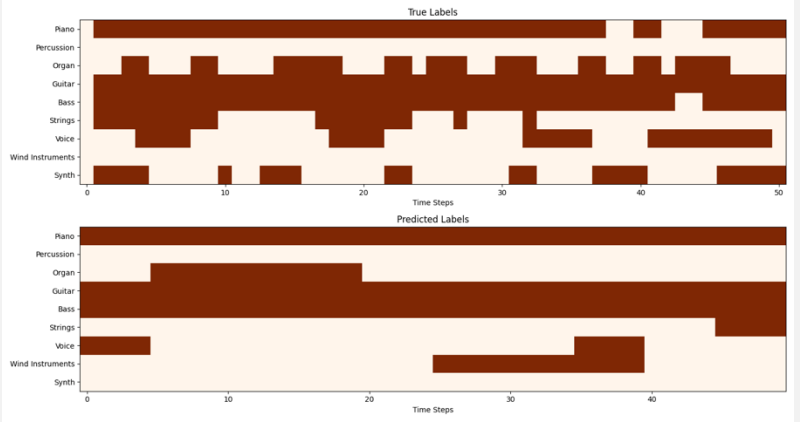
- **Track01876**



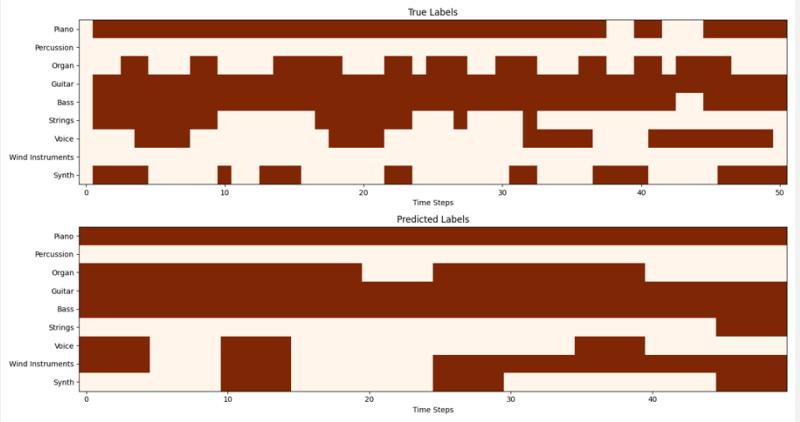
Vanilla



Weighted Loss



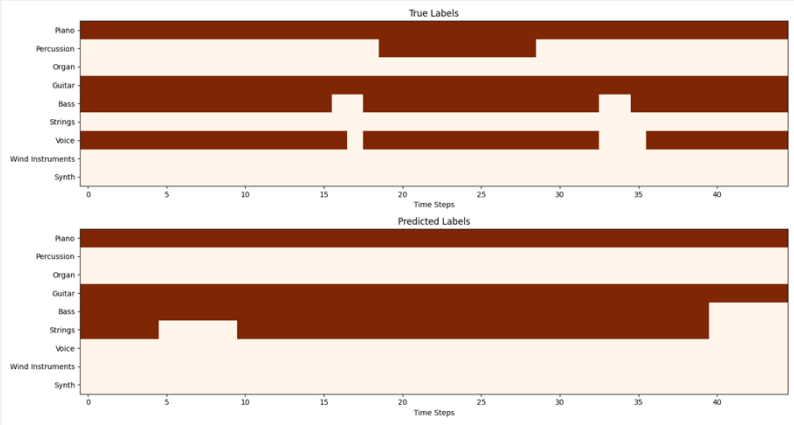
Fine-tuned



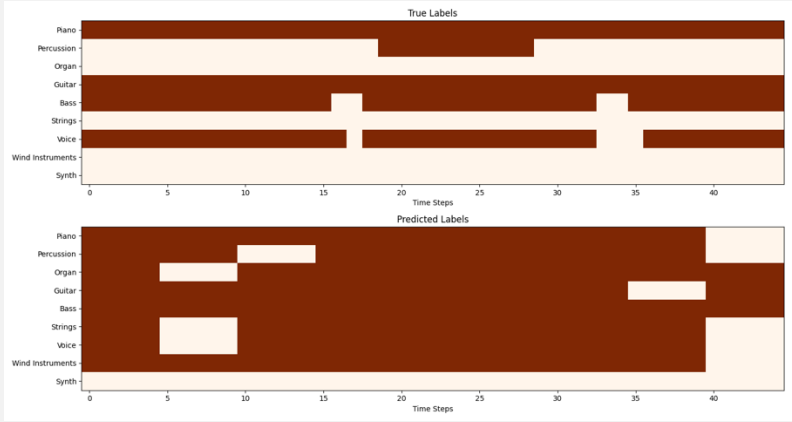
Fine-tuned + Weighted Loss

APPENDIX

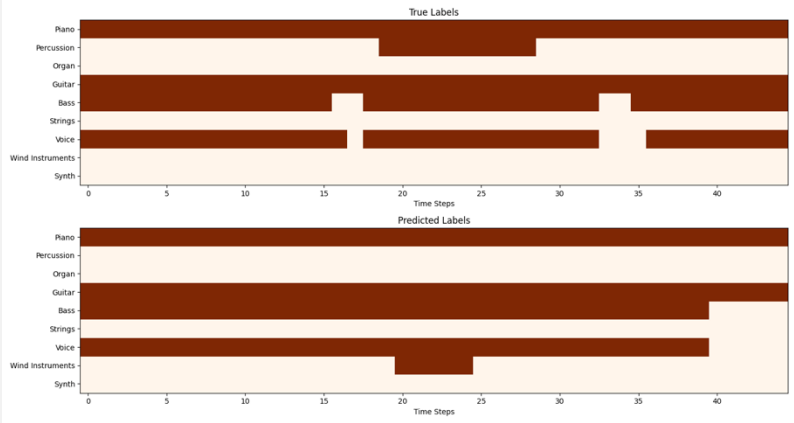
- **Track02100**



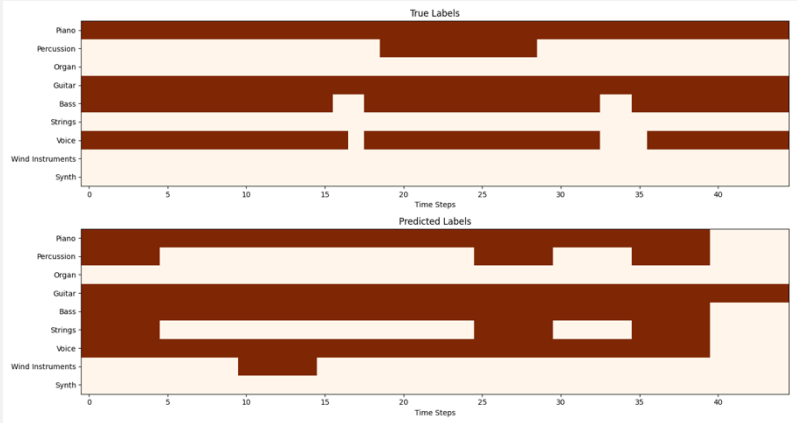
Vanilla



Regularization



Fine-tuned



Fine-tuned + Regularization